# Exponential and Logarithmic Equations

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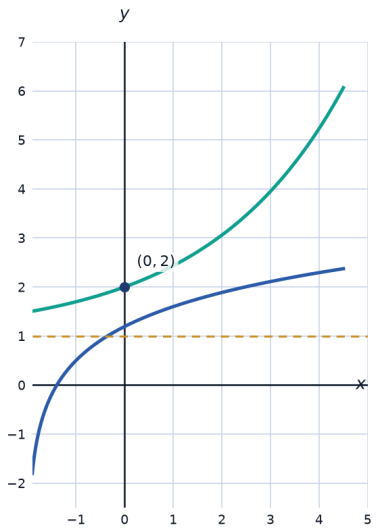
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Exponential equations

Solve $3^{x+1} = 81$ exactly.

**2** [Foundational] • Target: Exponential equations

Solve $5^{2x} = 17$. Give an exact answer and a decimal to three places.




**3** [Foundational] • Target: Logarithmic equations

Solve $\log_2 x + \log_2 (x - 6) = 4$, checking for extraneous solutions.

**4** [Foundational] • Target: Logarithmic equations

The pH of a solution is $\text{pH} = -\log[\text{H}^+]$. Find the hydrogen-ion concentration $[\text{H}^+]$ of a solution with pH 4.6, in scientific notation to two significant figures.

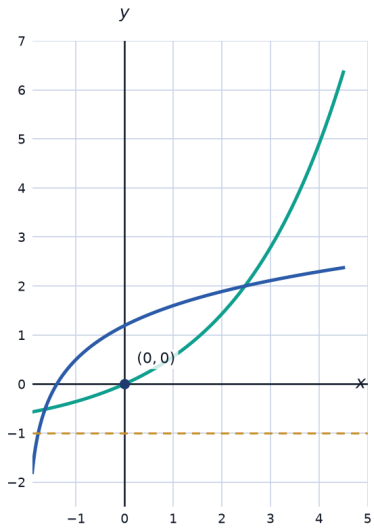
**5** [Foundational] • Target: More exponential equations

Solve $2^x = 3^{x-1}$, giving an exact expression and a decimal to three places.

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**6** [Foundational] • Target: More exponential equations

Solve $4^x - 3 \cdot 2^x - 4 = 0$ exactly, using the substitution $u = 2^x$.




**7** [Foundational] • Target: Systems and applications

A bacteria culture grows according to $N(t) = 500e^{kt}$. If the population reaches 2000 after 6 hours, find $k$ (4 d.p.) and hence the time for the population to reach 10,000, to one decimal place.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Systems and applications

Solve the equation $e^x + e^{-x} = 3$ exactly, using the substitution $u = e^x$ (this expression is related to $2\cosh x$).

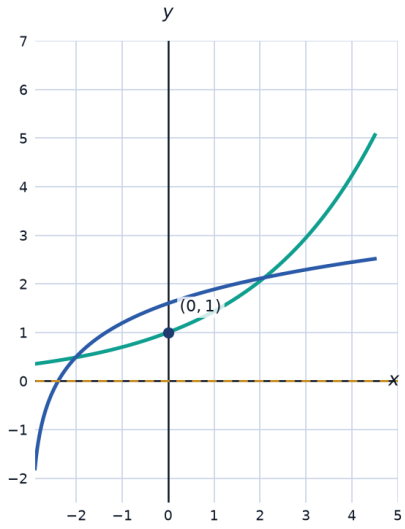
**9** [Intermediate] • Target: Combined logarithmic and exponential practice

Solve $9^x = 27^{x-1}$ exactly, writing both sides with base 3.

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**10** [Intermediate] • Target: Equations requiring both exponential and logarithmic techniques

Solve $e^{\ln x + \ln 2} = 10$ for $x$, simplifying the exponent first.




**11** [Intermediate] • Target: Verifying solutions numerically

For the equation $5^{2x} = 17$ solved earlier ($x \approx 0.880$), verify the solution by substituting back and computing both sides to 2 decimal places.

**12** [Intermediate] • Target: Equations with different bases on each side

Solve $10^{2x-1} = 500$, giving an exact logarithmic expression and a decimal to three places.

**13** [Intermediate] • Target: A modeling problem requiring both equation types

A population model is $P(t) = 800e^{0.05t}$. Find the time $t$ when $P(t) = 2000$, to two decimal places, and separately find $P^{-1}(1500)$ (the time when the population reaches 1500), to two decimal places.

**14** [Intermediate] • Target: Checking solutions by substitution

Verify the solution $x = 8$ found earlier for $\log_2 x + \log_2 (x - 6) = 4$ by substituting back into the original equation.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

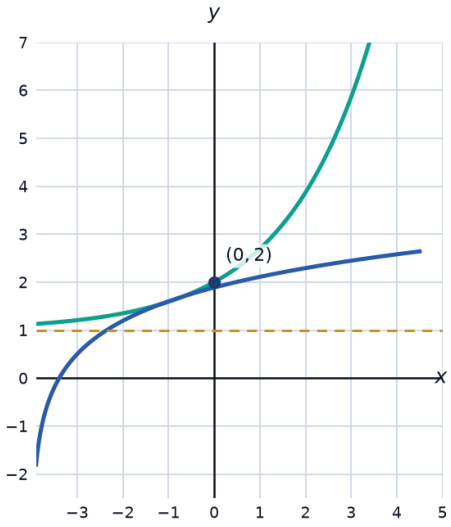
**15** [Challenging] • Target: Checking solutions by substitution • Cross-Lesson: 3.01 The Unit Circle and Exact Values

A radioactive sample decays according to $A(t) = A_0 e^{-0.08t}$. If the initial amount is 500 g, find the time $t$ (to two decimal places) at which only 50 g remains, and separately find the time at which 90% of the original sample has decayed away.

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**16** [Challenging] • Target: Equations requiring both exponential and logarithmic techniques • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Level 2 · MediumSolve the logarithmic equation $\log_2(x+3) + \log_2(x-1) = 5$ analytically for all real solutions. Show the verification step that eliminates extraneous roots.




**17** [Challenging] • Target: Verifying solutions numerically • Cross-Lesson: 3.03 Sum and Difference Formulas

Find all real solutions to the rational equation $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6}$.

**18** [Challenging] • Target: Equations with different bases on each side • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Solve the exponential equation $4^{x+1} - 5(2^x) + 1 = 0$ for all real values of $x$.

**19** [Challenging] • Target: A modeling problem requiring both equation types • Cross-Lesson: 4.01 Circles and Parabolas

Solve the logarithmic equation $\log_4(x+6) + \log_4(x-6) = 3$, checking for extraneous roots.

**20** [Challenging] • Target: Checking solutions by substitution • Cross-Lesson: 4.02 Ellipses

[Exponential & Logarithmic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Exponential equations

What is the sum of all real solutions to the equation $2^{\log_5(x^2)} - 6 \cdot x^{\log_5(2)} + 8 = 0$ ?

**a.** 30

**b.** 26

**c.** 6

**d.** 3

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**22** [Foundational] • Target: Logarithmic equations

The rate of change of a quantity $N$ is proportional to its current value, modeled by the differential equation $\frac{dN}{dt} = kN$, where $k$ is a constant. If the initial population at time $t = 0$ is $N_0$, and at time $t = 5$ years, the population has grown to $2N_0$, what is the population at time $t = 15$ years, expressed in terms of $N_0$?

**a.** $6N_0$

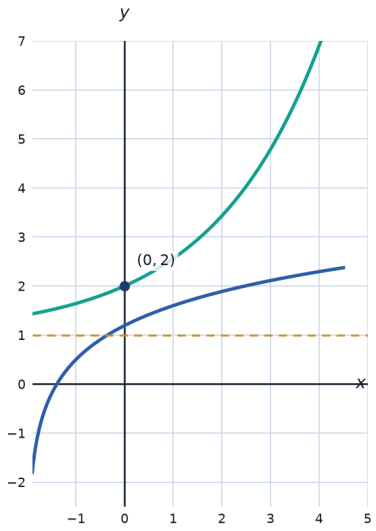
**b.** $8N_0$

**c.** $7N_0$

**d.** $10N_0$

**23** [Foundational] • Target: More exponential equations

The equation $\log_a x = b$ and $\log_b x = a$ simultaneously implies:




**a.** No simultaneous solution

**b.** $a = b$

**c.** $x = 0$

**d.** $x = a^b = b^a$

**24** [Foundational] • Target: Systems and applications

Find the number of solutions to $e^x = x + 2$.

**a.** 1

**b.** 2

**c.** Infinitely many

**d.** 0

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**25** [Foundational] • Target: Combined logarithmic and exponential practice

Solve $e^x = x + 2$ (number of solutions).

**a.** 0 solutions

**b.** 2 solutions (one for $x > 0$, one for $x < 0$)

**c.** 1 solution

**d.** Infinitely many

**26** [Foundational] • Target: Equations requiring both exponential and logarithmic techniques

Find all solutions to $4^x - 6 \cdot 2^x + 8 = 0$.

**a.** $x = 0, 1$

**b.** $x = 2, 4$

**c.** $x = 3, 4$

**d.** $x = 1, 2$

**27** [Foundational] • Target: Verifying solutions numerically

Find the number of real solutions to $\ln x = x^2 - 3$.

**a.** 3

**b.** 1

**c.** 0

**d.** 2

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**28** [Foundational] • Target: Equations with different bases on each side

The graph of the function $f(x) = a \cdot 2^{x-h} + k$ is shown below. The graph passes through the points $(2, 9)$ and $(4, 33)$. The horizontal asymptote is $y = 1$. Find the value of $a$.




**a.** 4

**b.** 2

**c.** 1/2

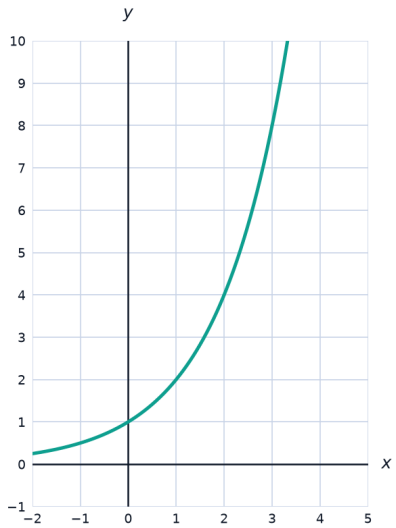
**d.** 8

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** **[Intermediate]** • **Target: A modeling problem requiring both equation types**

Consider the function $f(x) = 2^x$. Which of the following points lies on the graph of $f(x)$?




**a.** $(1, 1)$

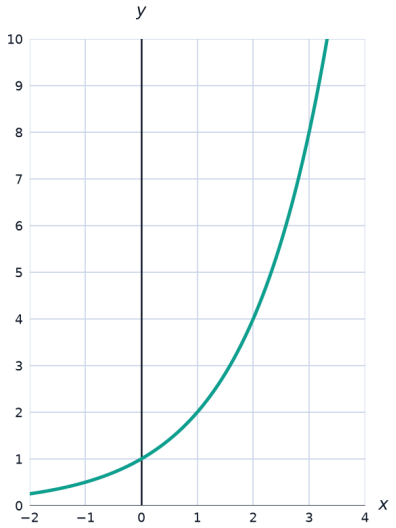
**b.** $(2, 4)$

**c.** $(3, 6)$

**d.** $(0, 2)$

**30** **[Intermediate]** • **Target: Checking solutions by substitution**

The graph of the exponential function $f(x) = 2^x$ is given. A point P on the graph has a y-coordinate of 8. What is the x-coordinate of point P?




**a.** 1

**b.** 2

**c.** 3

**d.** 4

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**31** **[Intermediate]** • **Target: Exponential equations**

The function $f$ is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k,$ where $k$ is a constant. The function $g$ is defined for $x \in \mathbb{R}$ by $g(x) = |x - 4|.$ Given that $f^{-1}(4) = 2$, find the value of $k$ and all real solutions to the equation $f^{-1}(x) = g(x)$.

**a.** $k = -2;\ x = \frac{5}{2},\ x = 7$

**b.** $k = -2;$ \; x = 7\)

**c.** $k = 2;\ x = \frac{7}{2},\ x = 5$

**d.** $k = -2;\ x = -\frac{5}{2},\ x = 7$

**32** **[Intermediate]** • **Target: Logarithmic equations**

The bar graph shows the distribution of the number of goals scored per game by a soccer team across all 40 games of its regular season. If the team plays $k$ post-season games and scores exactly 3 goals in each of those games, the mean number of goals scored per game across all regular-season and post-season games combined will be exactly 2. What is the value of $k$?

**a.** 8

**b.** 12

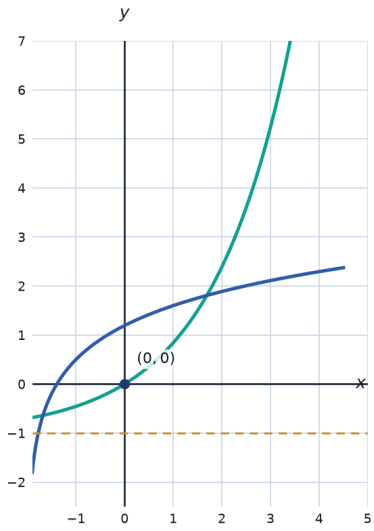
**c.** 20

**d.** 24

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**33** [Intermediate] • Target: More exponential equations

Consider the equation $\sqrt[3]{25 - x^2} + \sqrt[3]{x^2 + 3} = 4$ What is the product of all real solutions to this equation?




**a.** $-24$

**b.** $24$

**c.** $-48$

**d.** $0$

**34** [Intermediate] • Target: Systems and applications

Consider the function $f(x) = \log_3(ax + b)$, where $a$ and $b$ are positive constants. If the graph of $y = f(x)$ passes through the points $(1, 2)$ and $(10, 3)$, find the value of $a$.

**a.** 3

**b.** 1

**c.** 1/3

**d.** 9

**35** [Intermediate] • Target: Combined logarithmic and exponential practice

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?

**a.** $y = -5$

**b.** $y = 0$

**c.** $y = 3$

**d.** $x = -1$

**36** [Intermediate] • Target: Equations requiring both exponential and logarithmic techniques

The graph of the function $g(x) = a \cdot b^x$ passes through the points $(1, 6)$ and $(3, 24)$. If $a$ and $b$ are positive constants, what is the value of $a$?

**a.** 1.5

**b.** 2

**c.** 3

**d.** 6

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**37** [Intermediate] • Target: Verifying solutions numerically

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of $a$ and the value of $b$.

**a.** a = 4, b = 3

**b.** a = 3, b = 4

**c.** a = 6, b = 2

**d.** a = 2, b = 6

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Equations with different bases on each side • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of $a$.

**a.** 2

**b.** 3

**c.** 4

**d.** 6

**39** [Challenging] • Target: A modeling problem requiring both equation types • Cross-Lesson: 4.01 Circles and Parabolas

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(2, 36)$. If $b > 0$ and $beq1$, find the value of $a$.

**a.** 4

**b.** 3

**c.** 12

**d.** 1

**40** [Challenging] • Target: Checking solutions by substitution • Cross-Lesson: 4.02 Ellipses

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 6)$ and $(2, 18)$. What is the value of $b$?

**a.** 2

**b.** 3

**c.** 4

**d.** 6

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**41** [Challenging] • Target: Exponential equations • Cross-Lesson: 4.03 Hyperbolas

The graph of the exponential function $f(x) = a \cdot 2^x$ passes through the point $(3, 40)$. What is the value of $a$?

**a.** $\frac{5}{2}$

**b.** 5

**c.** 10

**d.** 20

**42** [Challenging] • Target: Logarithmic equations • Cross-Lesson: 5.01 Vectors: Components and Operations

Consider the matrix equation: $(3, -1; 2, k)(x; y) = (5; 7)$. If the matrix $(3, -1; 2, k)$ is singular, what is the value of $k$?

**a.** $-3/2$

**b.** $3/2$

**c.** $-2/3$

**d.** $2/3$

**43** [Challenging] • Target: More exponential equations • Cross-Lesson: 5.02 The Dot Product

What is the value of the exponential function $f(x) = 2^x$ when $x = 3$?

**a.** 6

**b.** 8

**c.** 9

**d.** 5

**44** [Challenging] • Target: Systems and applications • Cross-Lesson: 5.03 Parametric Equations

What is the value of $y$ when $x = 3$ for the function $f(x) = 2^{x+1}$?

**a.** 4

**b.** 8

**c.** 6

**d.** 16

**45** [Challenging] • Target: Combined logarithmic and exponential practice • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Solve: $e^{2x} = 5$. Express in terms of ln.

**a.** $x = \ln k**2 + 1 - k$

**b.** $x = \ln \frac{k**2 + 1}{k}$

**c.** $x = \frac{\ln k**2 + 1}{k}$

**d.** $x = \frac{k}{\ln k**2 + 1}$

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